



2025

M22

BOOKLET NO.

144026

Electrical Engineering
Paper – I

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

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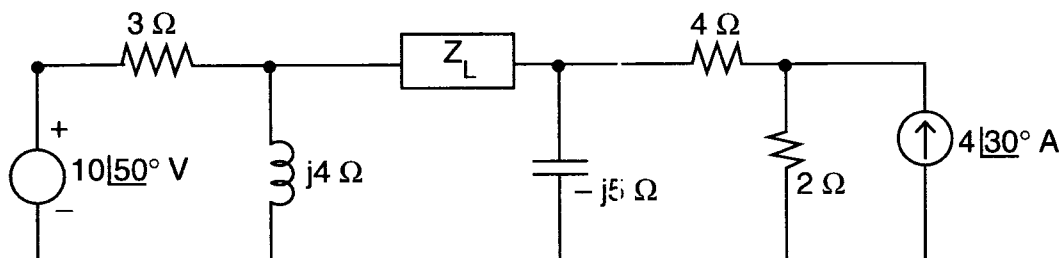


SECTION – A

Q1. Solve the following in about **150** words. **(5×10=50)**

- Q1. (a)** Two impedances, z_1 and z_2 are connected in parallel. The first branch takes a leading current of 16 A and has a resistance of 5Ω . The second branch takes a lagging current at power factor 0.8. The applied voltage is $(100 + j200)V$ and the total power supplied is 5 kW. Determine the values of circuit elements. **10**
- Q1. (b)** Determine causality for the input-output relation given below : **10**
 $y(t) = x(t) + 2x(3 - t).$
- Q1. (c)** Explain why Common Emitter (CE) configuration is most popular in transistor amplifier circuit. **10**
- Q1. (d)** Draw a 4-bit serial-in-serial-out shift register. Explain its working. **10**
- Q1. (e)** With the help of block diagram, explain the working of counter type ADC. **10**

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- Q2. (a)** For the network given in fig., find the value of impedance Z_L , which will draw maximum power from the network. Also find the value of maximum power drawn by Z_L . **20**

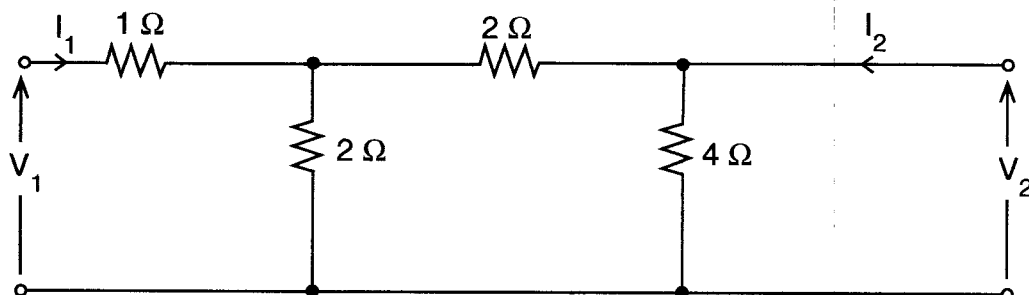


- Q2. (b)** Prove mathematically that the operating point in a potential-divider biasing circuit (of transistor) is independent of β (common emitter short circuit current gain). Make relevant assumptions. **20**
- Q2. (c)** Each arm of a three phase load comprises of a resistance of 25Ω , an inductance of 0.15 H and a capacitor of $120 \mu F$, connected in series. The supply is 415 V, 50 Hz. Calculate the total active power consumed when
- the three arms are connected in star.
 - the three arms are connected in delta.
- 10**
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- Q3. (a) Explain the properties of continuous time LTI system and discrete time LTI system. 20
- Q3. (b) What is race-around condition in J-K flip flop ? With the help of necessary diagram and truth-table, explain, how it is avoided in J-K master-slave flip-flop. 20
- Q3. (c) With the help of necessary diagrams, explain, how an OPAMP can be used as an integrator. 10

- Q4. (a) Find h-parameters (hybrid parameters) of the network shown in fig. Using h-parameters, determine, whether this two-port network is symmetric or not. 20



- Q4. (b) With the help of suitable diagram, explain the working of tuned-collector LC oscillator. 20
- Q4. (c) Determine the Z-transform and ROC (Region of Convergence) of the discrete time signal $x(n) = \{3, 2, 5, 7\}$. 10

SECTION – B

- Q5. Solve the following in about 150 words. (5×10=50)
- Q5. (a) The electric field amplitude of a uniform plane wave propagation in $\hat{a}_z$ direction is 250 V/m. If $\vec{E} = E_x \hat{a}_x$ and $\omega = 1.00$ Mega-rad/s, find (i) the frequency (ii) the wavelength (iii) amplitude of H. 10
- Q5. (b) A 200 V dc shunt motor is taking a current of 70 A and runs at 500 rpm. The resistance of the shunt field is 100 Ω and of the armature is 0.2 Ω . Find the value of resistance, that must be placed in series with the armature in order that the speed should be reduced to 350 rpm, the current in the armature remaining the same. 10
- Q5. (c) Draw and explain the steady-state characteristics of MOSFET. 10



Q5. (d) Let X be a continuous random variable with probability density function

$$f_X(x) = \begin{cases} x^2 \left(2x + \frac{3}{2} \right) & 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}, \text{ if } Y = \frac{2}{x} + 3, \text{ find var } (Y). \quad 10$$

Q5. (e) Explain discrete random variable and continuous random variable. 10

Q6. (a) In the region where $\sigma = 0$, $\epsilon = 2.5\epsilon_0$ and $\mu = 10\mu_0$, determine, whether following pair of fields satisfy Maxwell's equation or not.

$$\vec{D} = (z + 6 \times 10^7 t) \hat{a}_x, \vec{B} = (-754z - 4.52 \times 10^{10} t) \hat{a}_y \quad 20$$

Q6. (b) Why do we need a starter for starting a three-phase induction motor? With the help of diagram, explain star-delta starter used for starting 3-phase induction motor. 20

Q6. (c) Explain principle of operation of a thyristor. 10

Q7. (a) With the help necessary diagrams and equations, explain the working of step-down chopper (Buck-converter). 20

Q7. (b) Define Phase Modulation (PM) and Frequency Modulation (FM). Explain direct and indirect method of detection of FM wave. 20

Q7. (c) A 5 kVA, 400/200 V, 50 Hz, single-phase transformer gave the following results during open circuit and short circuit tests. 10

OC test : 400 V, 1 A, 60 W (hv side)

SC test : 15 V, 12.5 A, 50 W (hv side)

Calculate :

i) No load resistance (R_0) and magnetising reactance (X_m).

ii) Equivalent resistance (R_{01}) and equivalent reactance (X_{01}) referred to primary (hv) side.

Q8. (a) A transmission line has series impedance $Z = (R + j\omega L)\Omega$ and shunt admittance $Y = (G + j\omega C)\Omega$. Derive expression for voltage and current waves, in exponential form. 20

Q8. (b) With the help of diagram and voltage and current waveforms, explain single-phase bridge converter. 20

Q8. (c) Given two dielectric mediums, medium 1 is free space and medium 2 has $\epsilon_2 = 4\epsilon_0$, $\mu_2 = \mu_0$. Determine reflection coefficient, if $\theta_i = 30^\circ$, for perpendicular polarization. 10